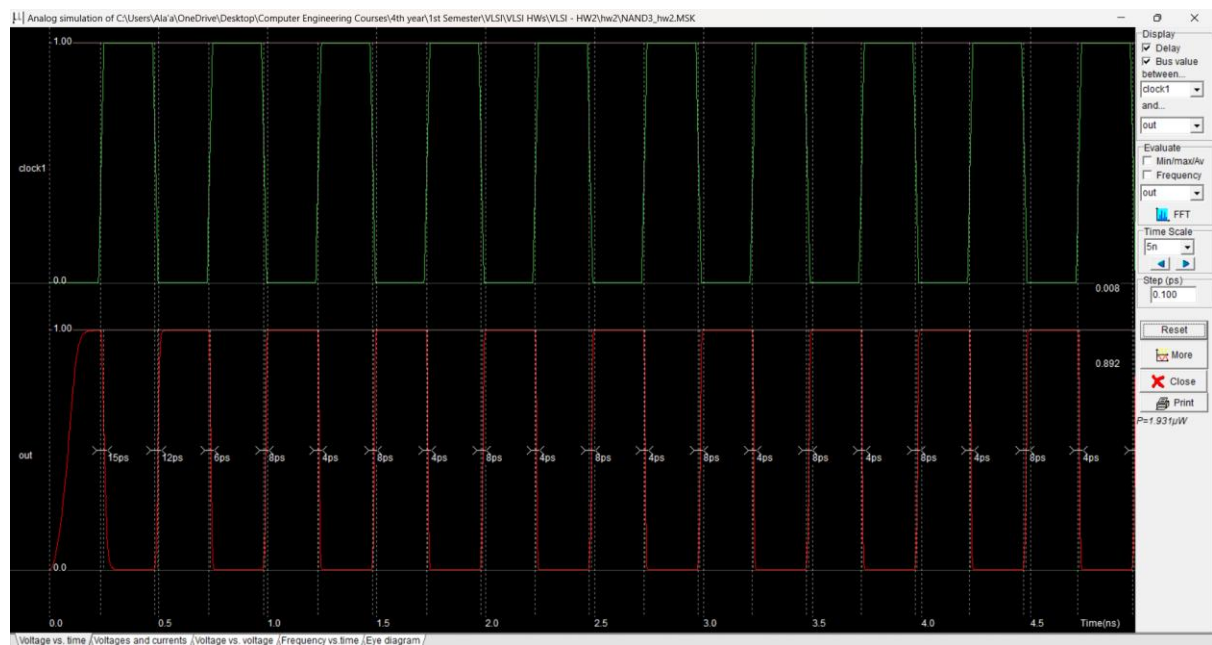
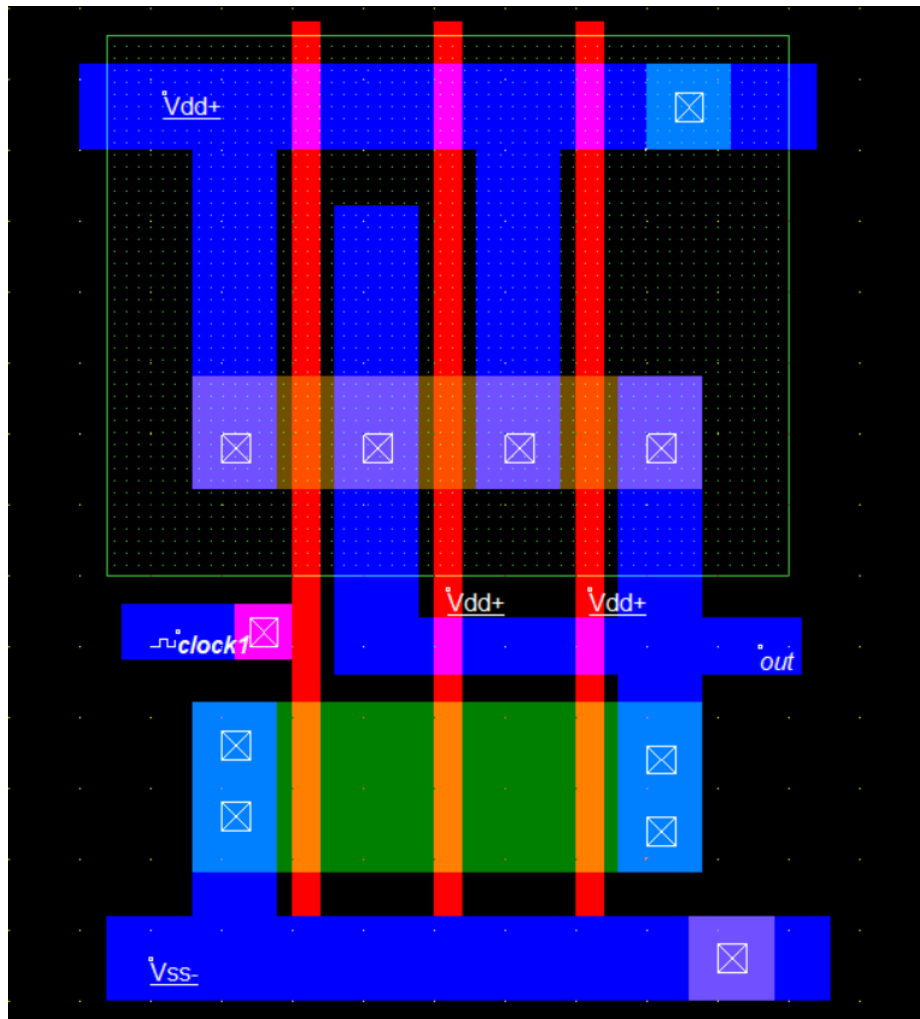
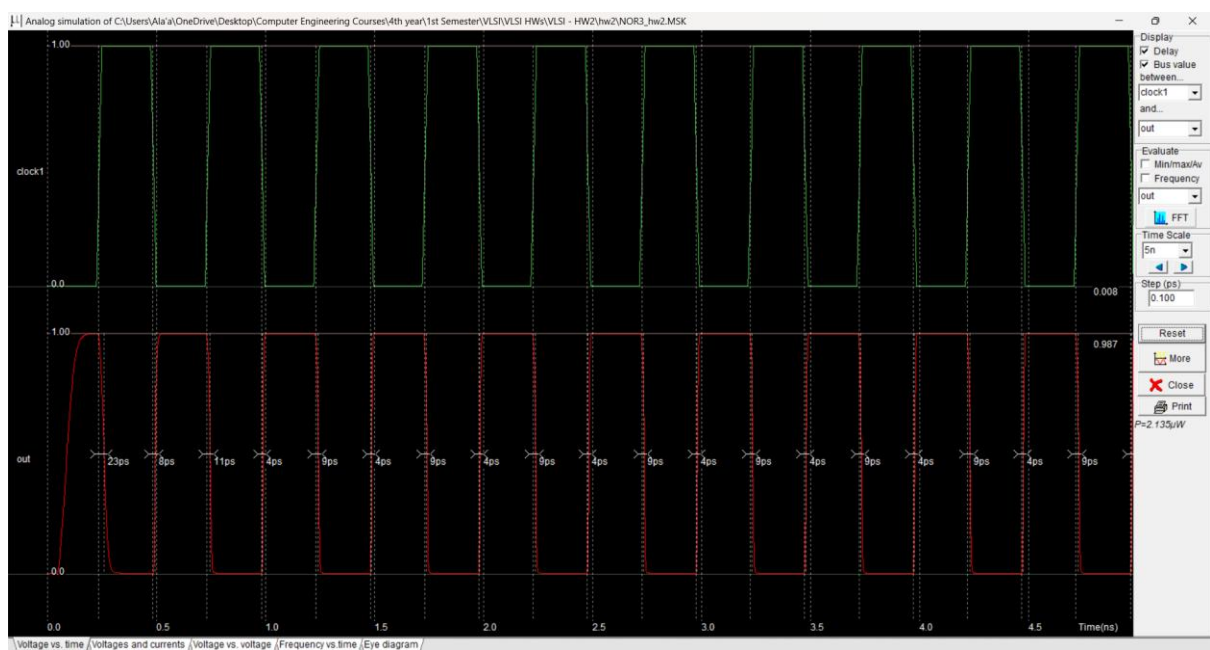
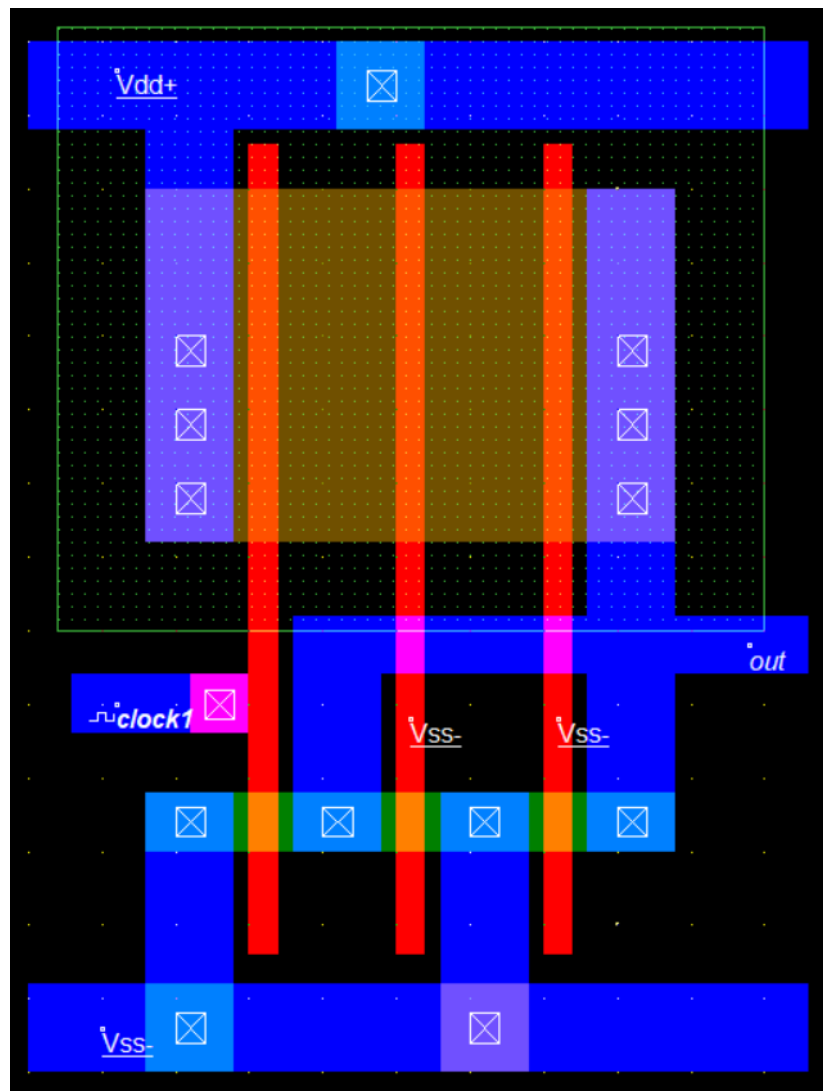


➤ First: The Implementation of The Circuit using Microwind:

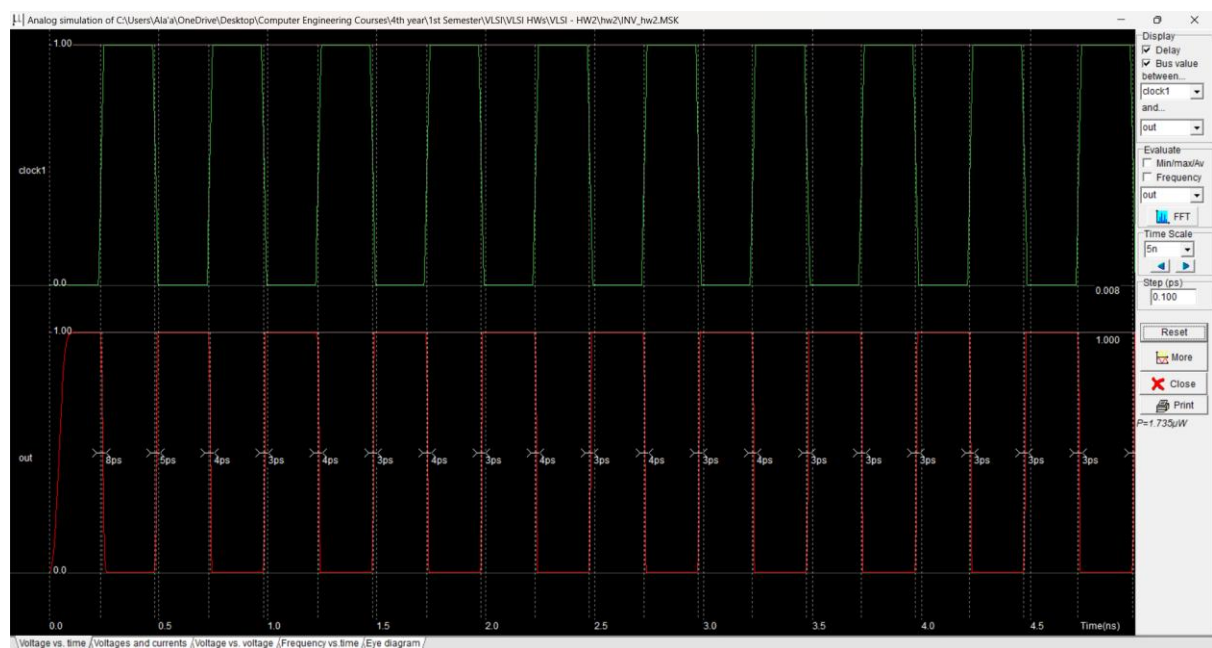
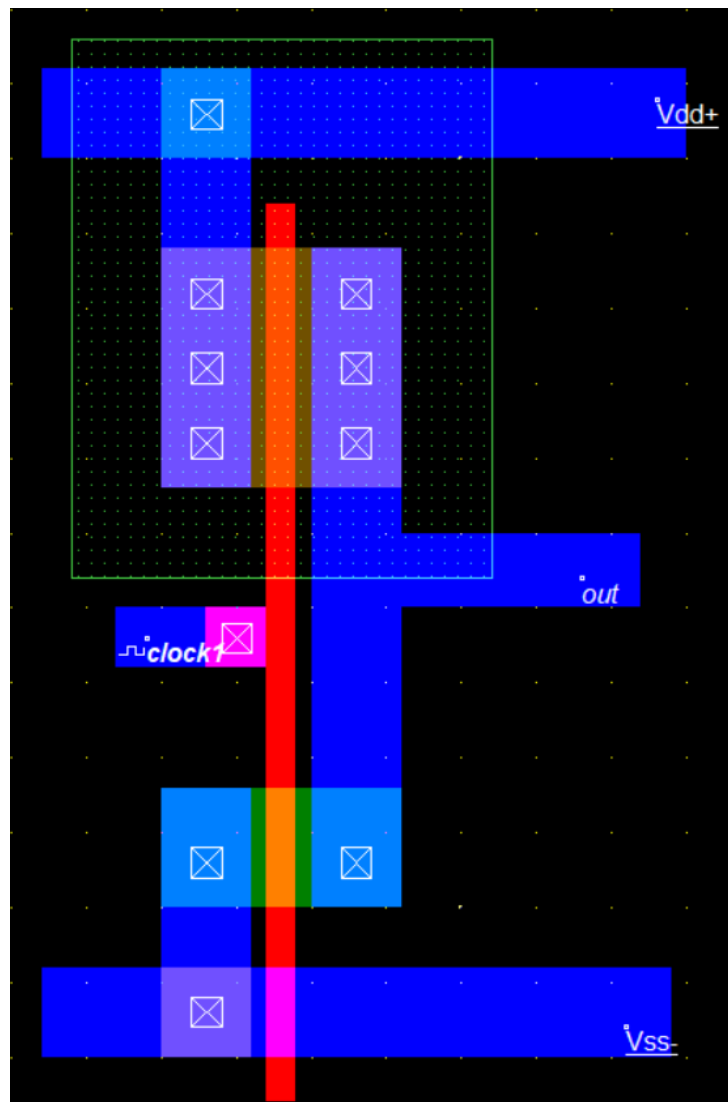
1. Create the NAND3_hw2.MSK gate:



2. Create the NOR3_hw2.MSK gate:

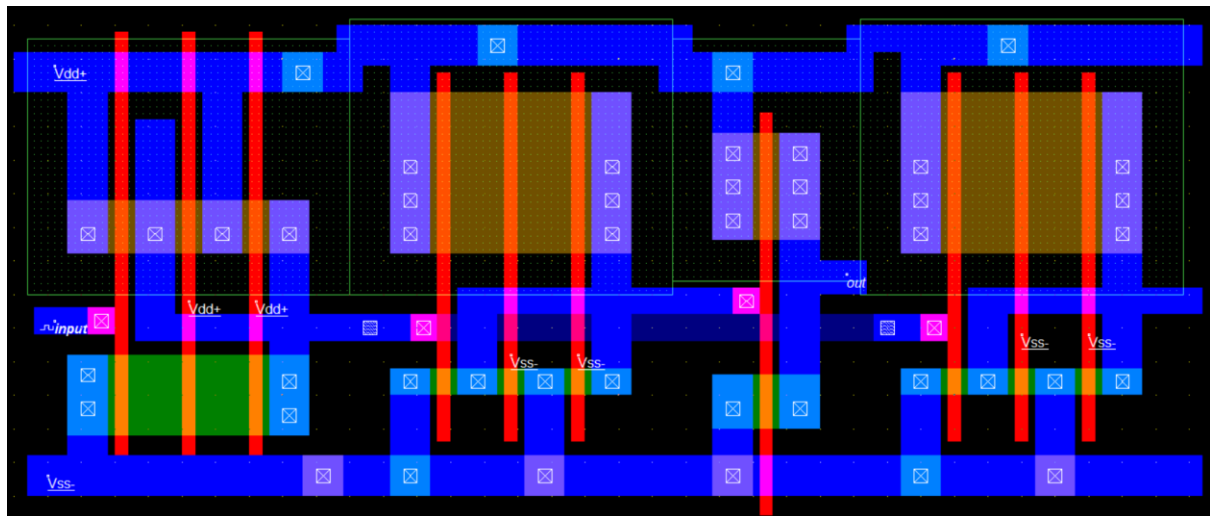


3. Create the INV_hw2.MSK gate:

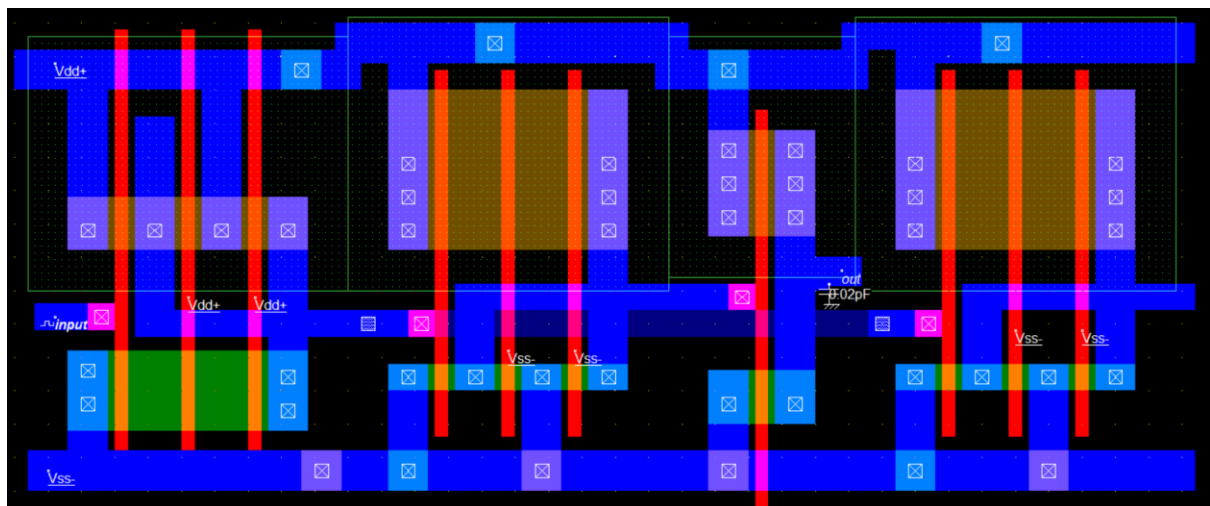


4. Create the layout for the Circuit and call it hw2.MSK:

a. With no load:

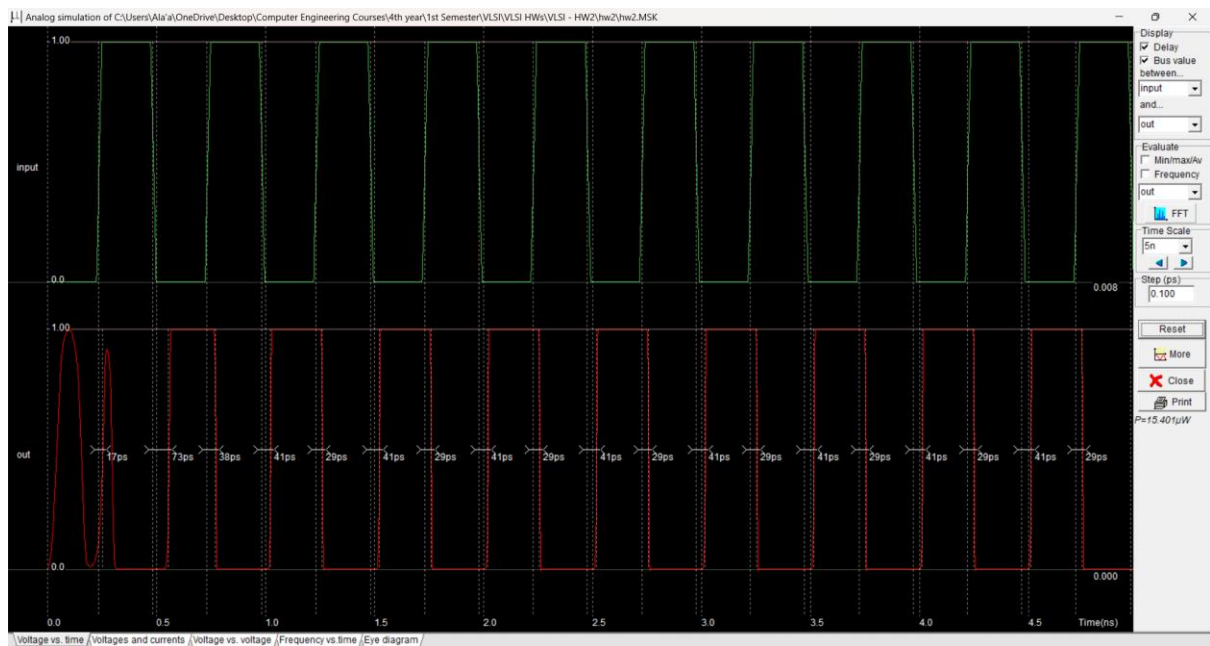


b. With load:

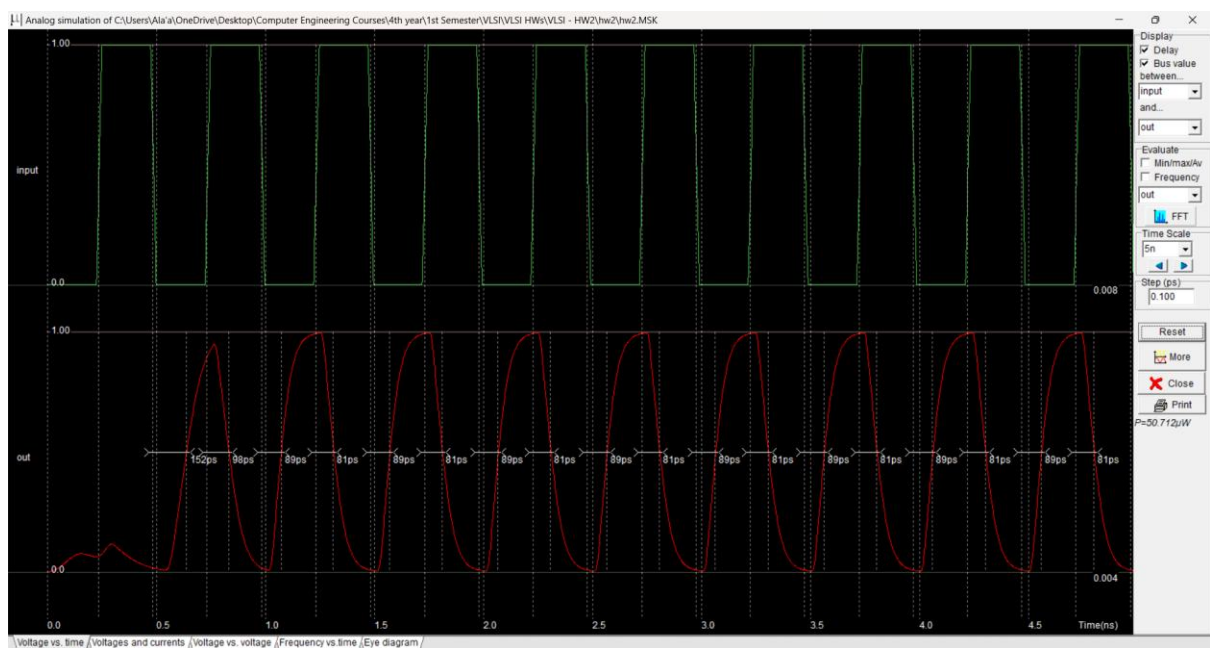


5. Simulate the Circuit and find out the delay between the Input and the output:

a. With no load:

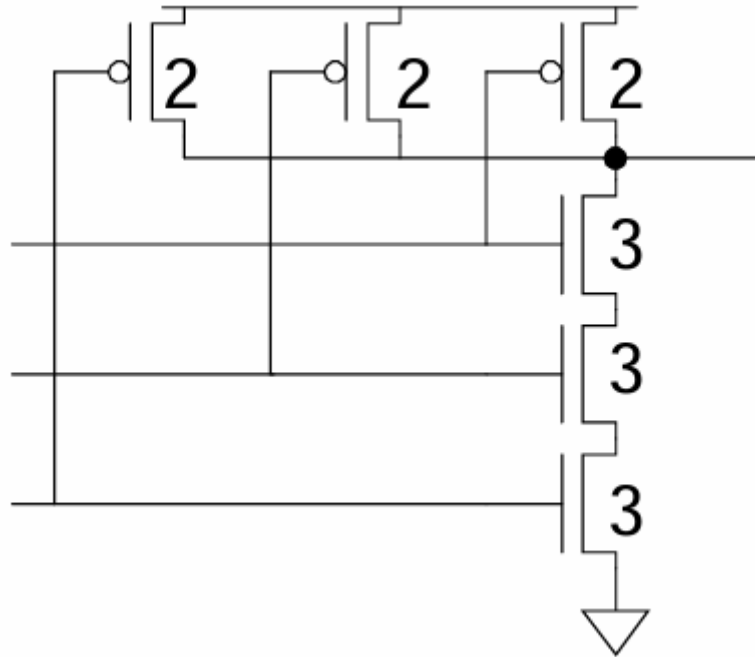


b. With load:



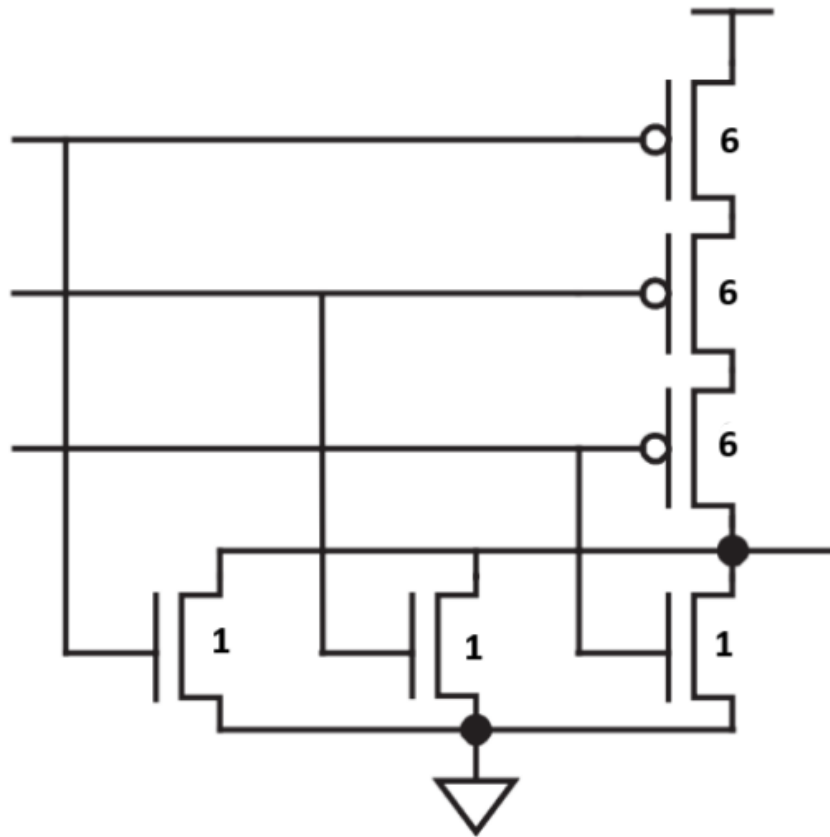
➤ **Second: Using the Logical Effort Model to Compute Manually the Delay of The Circuit:**

1. 3-Input NAND gate:



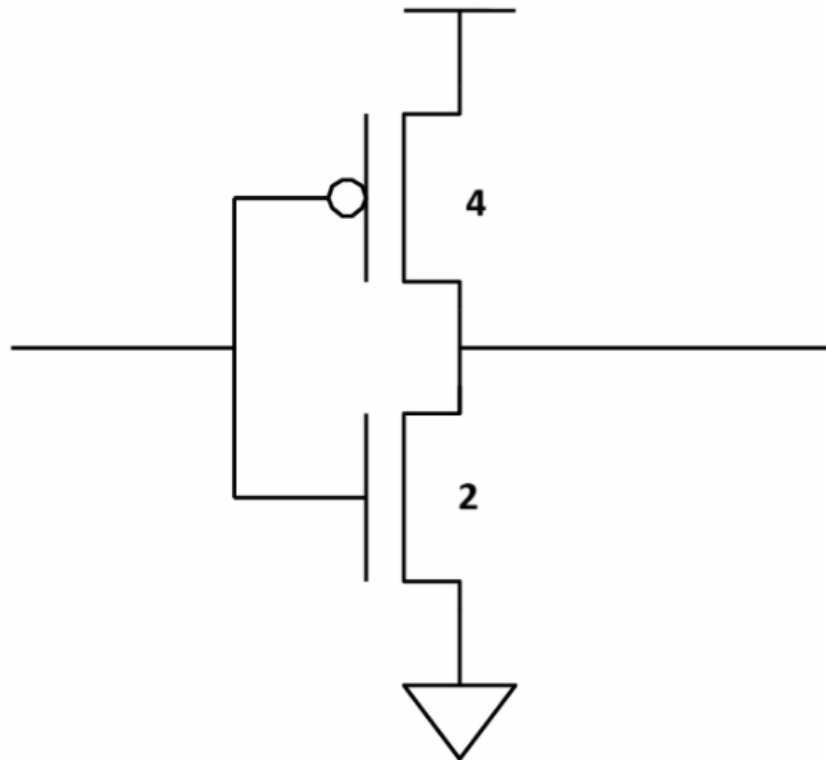
- W/L for N-MOS is $12/2$.
- W/L for P-MOS is $8/2$.
- $g = C_{in} / C_{in-inv} = 5/3$.
- $p = C_{out} / C_{out-inv} = (3+2+2+2)/3 = 3$.
- $h = h = C_{Load} / C_{in} = 2 * (7/5)$.

2. 3-Input NOR gate:



- W/L for N-MOS is $4/2$.
- W/L for P-MOS is $24/2$.
- $g = C_{in} / C_{in-inv} = 7/3$.
- $p = C_{out} / C_{out-inv} = (6+1+1+1)/3 = 3$.
- $h = C_{Load} / C_{in} = 6/7$.

3. Inverter gate:



- W/L for N-MOS is $8/2$.
- W/L for P-MOS is $16/2$.
- $g = C_{in} / C_{in-inv} = 1$.
- $p = C_{out} / C_{out-inv} = 1$.
- $h = C_{Load} / C_{in} = 21/6$.

Total delay = $d1 + d2 + d3$. (Note: $d = g * h + p$)

- $d1 = (5/3) * (2 * (7/5)) + 3 = 7.67$.
- $d2 = (7/3) * (6/7) + 3 = 5$.
- $d3 = 1 * (21/6) + 1 = 4.5$.

Total delay = 17.17 unit.

In 65 nm = $17.17 * 3ps = 51.51 ps$.

In a 600 nm = $17.17 * 60ps = 1030.2 ps$.